# Atlas2350 Rev A - PCBWay TURNKEY

Quote an initial lot of five completely assembled development boards.
PCBWay fabricates the PCB, supplies all exact-MPN BOM parts and assembles SMT + THT.
115 real components per board; 53 BOM groups; no DNP or customer-fit components.

**FABRICATION**

80 x 75 mm chamfered outline, four copper layers, finished thickness 1.6 mm.
Lead-free FR-4, ENIG, green soldermask, white silkscreen; IPC Class 2 workmanship.
Outer copper 1 oz, inner copper 0.5 oz; nominal dielectrics 0.12/1.255/0.12 mm.
Actual cured stackup/material details require quotation; no impedance certification claimed.
F/B: signals + GND. In1/In2: GND pours with documented signal escape corridors.
Minimum track / clearance: 0.150 / 0.127 mm. Hole-to-copper: >=0.178 mm.
Routing vias: 0.600 mm lands / 0.300 mm finished holes, 0.150 mm annular ring.
ALL 334 holes in drill/filled-capped-hole-schedule.csv require
IPC-4761 Type VII resin filling and copper capping, including 12 U6 thermal PTHs.
DO NOT fill connector component-lead holes. J6 finished holes are 0.700 mm.
Keep all listed NPTH mechanical holes non-plated. Excellon slots are routed.
Bare-board 100% electrical test against Atlas2350.ipc (IPC-D-356).

**ASSEMBLY / ORIENTATION**

Assemble every BOM row, including THT J5/J6/J7 and USB shell stakes.
Use exact packages: U2 wide SOIC8, U3 narrow SOIC8, U5 exact RNM0015A.
U5 pads9/11 are SWITCH NODES, not ground; there is no exposed pad16.
Use WS2812B-2020-V6 exactly. Do not substitute older LED logic-threshold variants.
Top paste Gerber supplied; PCBWay reconciles stencil/profile with component datasheets.
Inspect fine-pitch QFN / RNM / module thermal solder joints, including X-ray as needed.
Pick-and-place centroids / CAD anchors / pin1 occurrences are separately identified.
Reconcile reel zero angles, vacuum pick points and panel offsets before placement.

**MECHANICAL / HANDLING**

U6 antenna extends about 6.1 mm beyond the top board edge; no copper under antenna.
Panel rails, clamps and tabs must clear the module and its antenna keepout.
Preserve connector access at board edges and microSD insertion/ejection clearance.
No screws, battery, cable, microSD card or mating adapters are assembly components.

**ELECTRICAL / QUALIFICATION**

Battery interface is CUSTOM: J2 pin1 BAT+, pin2 GND, pin3 cell-mounted 10k NTC.
External protected 1S 4.2V LiPo >=500 mAh; no battery installed/shipped in this package.
No application or ESP-AT firmware image supplied; IC programming is not in this order.
Firmware controls USB100/500/suspend and radio enable. Default radio is disabled.
External RP core LDO: firmware must set VREG.HIZ and avoid low-power VREG mode.
U10 switches LED_VDD with SW5; RGB not guaranteed below LED_VDD 3.3V.
Blank RGB around VBAT 3.6V under load; the main rail remains 3.296V.
Unbuilt prototype: electrical/thermal/USB/RF qualification still required on first articles.
Clean ERC/DRC and pin checks are design checks, not factory DFM or functional approval.

Read fabrication-notes.md, assembly-notes.md and test-plan.md before quotation.